

PROGRAM: BANKER'S ALGORITHM

C PROGRAM

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, m, i, j, k;
```

```
    int allocation[10][10], max[10][10];
```

```
    int need[10][10];
```

```
    int available[10];
```

```
    int finish[10] = {0};
```

```
    int safeSeq[10];
```

```
    int count = 0;
```

```
    printf("Enter Number of Processes: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter Number of Resources: ");
```

```
    scanf("%d", &m);
```

```
    printf("\nEnter Allocation Matrix:\n");
```

```
    for (i = 0; i < n; i++) {
```

```
        for (j = 0; j < m; j++) {
```

```
            scanf("%d", &allocation[i][j]);
```

```
        }
```

```
    }
```

```
    printf("\nEnter Maximum Matrix:\n");
```

```
    for (i = 0; i < n; i++) {
```

```
        for (j = 0; j < m; j++) {
```

```
            scanf("%d", &max[i][j]);
```

```
        }
```

```
    }
```

```
    printf("\nEnter Available Resources:\n");
```

```

for (i = 0; i < m; i++) {
    scanf("%d", &available[i]);
}

for (i = 0; i < n; i++) {
    for (j = 0; j < m; j++) {
        need[i][j] = max[i][j] - allocation[i][j];
    }
}

while (count < n) {
    int found = 0;
    for (i = 0; i < n; i++) {
        if (finish[i] == 0) {
            for (j = 0; j < m; j++) {
                if (need[i][j] > available[j])
                {
                    break;
                }
            }
            if (j == m) {
                for (k = 0; k < m; k++)
                {
                    available[k] += allocation[i][k];
                }
                safeSeq[count++] = i;
                finish[i] = 1;
                found = 1;
            }
        }
    }
}

```

```

    if (found == 0) {
        printf("\nSystem is NOT in Safe State\n");
        return 0;
    }
}

printf("\nSystem is in Safe State\n");
printf("Safe Sequence: ");
for (i = 0; i < n; i++) {
    printf("P%d ", safeSeq[i]);
}

printf("\n");
return 0;
}

```

SHELL SCRIPT :

```

GNU nano 8.7.1
#!/bin/bash

echo "Enter Number of Processes:"
read n

echo "Enter Safe Sequence (space separated):"
read -a seq

echo "Safe Sequence is:"

for ((i = 0; i < n; i++))
do
    echo -n "P${seq[$i]} "
done

echo

echo "System is in Safe State"

```

Output :

```
(ganesh@LAPTOP-R3RDH9NG)~  
$ nano bankers.sh  
  
(ganesh@LAPTOP-R3RDH9NG)~  
$ chmod +x bankers.sh  
  
(ganesh@LAPTOP-R3RDH9NG)~  
$ bash bankers.sh  
Enter Number of Processes:  
3  
Enter Safe Sequence (space separated):  
12  
Safe Sequence is:  
P12 P P  
System is in Safe State
```